Executive Summary

This project analyses a retail sales dataset to identify the key drivers of revenue and profitability across products, regions, and customer segments. SQL Server was used to perform data cleaning, data modelling, and analytical querying. A relational schema was designed to separate customers, orders, products, and order line items to enable efficient analysis.

Power BI was used to develop an interactive dashboard that visualises key business metrics including sales performance, profitability by product category, regional performance, customer contribution, and the relationship between discounting and profit.

The analysis revealed that Technology products generate the highest revenue, while certain sub-categories such as Tables and Bookcases generate negative profit. The West region produces the strongest sales performance, and higher discount levels correlate with declining profitability.

Based on these findings, several strategic recommendations are proposed to improve margin performance and optimise discount strategies.

Business Problem

The organisation seeks to understand which products, regions, and customer segments contribute positively to profitability and where margin erosion is occurring due to discounting or operational inefficiencies.

The goal of this project is to identify the key drivers of revenue and profitability and provide actionable insights for improving business performance.

Analytical Approach

To address the business problem, the analysis focused on four key analytical dimensions that influence retail performance.

1. Product Performance Analysis

This analysis evaluates which product categories and sub-categories generate the highest revenue and profit.

Understanding product performance helps identify:

- high-performing product groups
- underperforming or loss-making products
- opportunities for product portfolio optimisation.

2. Regional Sales Performance

Regional analysis examines how sales and profitability vary across geographic regions.

This helps determine:

- which regions generate the most revenue
- where market performance is weaker
- where expansion or marketing investment may be required.

3. Customer Contribution Analysis

Customer analysis identifies the customers contributing the highest sales value.

This enables the business to:

- recognise high-value customers
- prioritise customer retention strategies
- evaluate customer concentration risk.

4. Discount Impact Analysis

Discount analysis evaluates how different discount levels influence overall profitability.

This helps determine:

- whether discounting drives revenue growth
- whether high discount levels erode profit margins.

The insights from these analytical areas were used to design the SQL queries and Power BI dashboard visualisations.

Stakeholders

The insights generated from this analysis would be valuable to several business stakeholders including:

Sales Management

To identify high-performing product categories, regions, and key customers contributing to revenue.

Pricing and Promotions Teams

To evaluate how discount levels impact profitability and determine whether current discount strategies are sustainable.

Product Management

To identify underperforming product lines and assess whether pricing, sourcing, or inventory strategies should be adjusted.

Executive Leadership

To monitor overall business performance and identify opportunities for revenue growth and margin improvement.

Dataset Overview

The dataset used in this project is the Superstore retail dataset, which contains transactional sales data including customer information, product details, order information, and financial metrics.

Key variables include:

- Customer_ID
- Order_ID
- Product_ID
- Sales
- Quantity
- Discount
- Profit
- Region
- Order Date
- Sub-Category

The dataset represents transaction-level data where each row corresponds to a product line item within an order.

Data Cleaning & Data Integrity Documentation

Data Quality Assessment Overview

A total of 3 date inconsistencies and 1 financial datatype issue were identified and resolved.

Prior to performing analytical queries and dashboard development, a structured data quality assessment was conducted to ensure the dataset was reliable for business analysis.

The review focused on:

- Missing values

- Datatype inconsistencies
- Transaction-level consistency
- Financial field integrity

Two issues were identified during this process.

Issue 1 – Missing Order Date Values

Issue Identification

During validation checks, three (3) records were identified with missing values in the Order Date field.

Initial observations showed that:

- Each affected record shared the same Order ID as other rows
- The corresponding Ship Date was populated
- The rows represented line items within the same transaction

This indicated the issue was likely due to a data entry or export inconsistency rather than missing transactional data.

Risk Assessment

If left unresolved, missing Order Date values would have:

- Distorted time-based analysis (monthly trends, year-over-year comparisons)
- Impacted shipping lead-time calculations
- Reduced accuracy in temporal aggregations

Resolution Approach

To preserve transactional integrity:

1. Records were grouped by Order ID
2. Valid Order Date values within the same transaction group were identified
3. Missing Order Date values were imputed using the confirmed date from the same transaction

This approach ensured:

- Logical consistency across all line items within a single order
- Alignment between Order Date and Ship Date
- Accurate time-series reporting downstream

2.4 Post Validation

After correction:

- All records contained valid Order Date values
- No inconsistencies remained within grouped transactions
- Date relationships were logically consistent

Issue 2 – Profit Column Datatype Validation Failure

Issue Identification

During the import of the dataset into SQL Server:

- One value in the Profit column failed datatype validation
- SQL Server automatically replaced the value with NULL

This was identified through validation queries checking for unexpected NULL values in financial columns.

Risk Assessment

A NULL value in the Profit column would have:

- Impacted aggregate profit calculations
- Skewed profit margin metrics
- Reduced reliability of financial reporting outputs

Resolution Approach

The following steps were taken:

1. A query was executed to identify rows where Profit IS NULL
2. The affected record was isolated
3. The original dataset was referenced to verify the correct value
4. An UPDATE statement was executed to restore the accurate profit value

This ensured:

- Financial data integrity
- Accurate aggregation and KPI reporting
- Consistency between source and database values

Data Integrity Confirmation

Ensuring data integrity was critical to accurately assessing profitability drivers and margin erosion.

Following resolution of identified issues:

- All date fields were complete and transactionally consistent

- Financial columns contained no unintended NULL values
- Dataset integrity was confirmed prior to analytical modelling
- The cleaned dataset was used for all downstream SQL analysis and Power BI reporting

Data Model Design

To improve analytical performance and eliminate redundancy, the dataset was normalised into four relational tables.

1) Customers

Purpose: Store customer details once.

Key: Customer_ID (natural key from dataset)

Columns:

- Customer_ID (PK)
- Customer_Name
- Segment

2) Products

Purpose: store product details once.

Key: Product_ID (natural key from dataset)

Columns:

- Product_ID (PK)
- Product_Name
- Category
- Sub_Category

3) Orders

Purpose: store order "header" info (one row per Order_ID).

Key: Order_ID (PK)

Columns:

- Order_ID (PK)
- Order_Date
- Ship_Date
- Ship_Mode
- Customer_ID (FK → Customers)

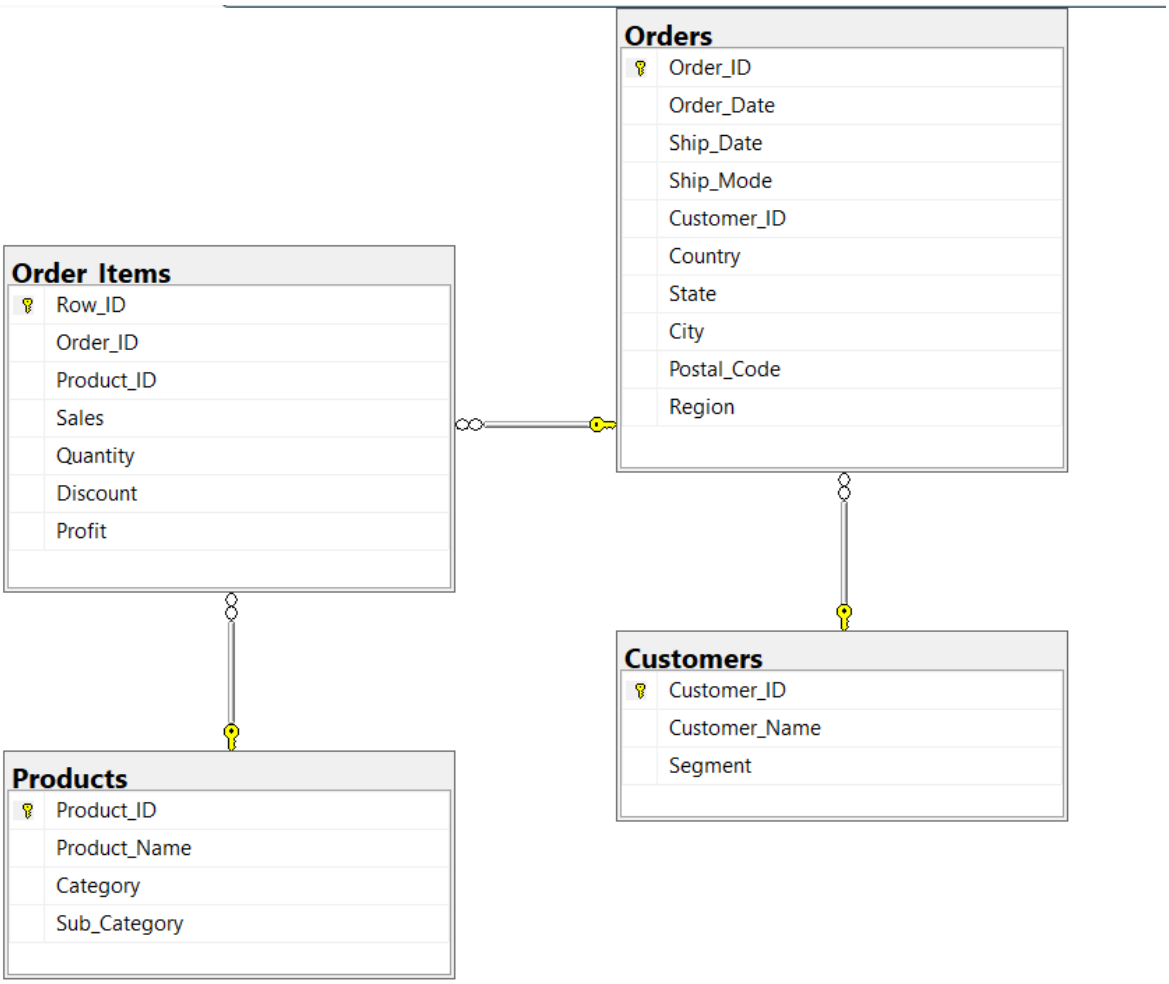
- Country
- State
- City
- Postal_Code
- Region

4) Order_Items

Purpose: store line items (one row per product line in an order).

Columns:

- Row_ID (PK)
- Order_ID (FK → Orders)
- Product_ID (FK → Products)
- Sales
- Quantity
- Discount
- Profit



Customers to Orders (1 to Many) as one customer can place many orders

Orders – OrderItems (1 to Many) as one order can contain many items

Products – OrderItems (1 to Many) as one product can appear in many orders

Data Transformation and Dimension Handling

Product Dimension Deduplication

Issue Identification

During the population of the Products dimension table, a Primary Key violation occurred when inserting records from the source dataset.

This occurred because the source dataset is structured at the order line level, meaning the same Product_ID appears multiple times across different transactions.

As a result, attempting to insert all rows directly into the Products table caused duplicate Product_ID values, violating the table's primary key constraint.

Resolution Approach

To resolve this, the dataset was transformed so that only one record per Product_ID was inserted into the Products table.

This was achieved by grouping the source data by Product_ID and selecting a single representative record for each product.

I used the following query:

```
INSERT INTO dbo.Products (Product_ID, Product_Name, Category, Sub_Category)
SELECT
Product_ID,
MAX(Product_Name),
MAX(Category),
MAX(Sub_Category)
FROM dbo.[Sample - Superstore -transformed]
GROUP BY Product_ID;
```

This ensured that:

- Each product appears **once** in the Products table
- The primary key constraint is preserved
- Redundant product data is removed
- The table structure supports clean relational joins with Order_Items

Product Attribute Standardisation

During the creation of the Products dimension table, it was observed that some Product_ID values were associated with slightly inconsistent descriptive attributes (such as product names or sub-categories) across different transaction records.

This occurs because the source dataset is transactional and may contain minor variations in product descriptions.

To ensure each Product_ID corresponds to a single record in the Products table, a grouping strategy was applied so that one consistent set of attributes was selected for each product.

This ensured that the dimension table maintained a one-to-one relationship between Product_ID and product attributes, preserving referential integrity for downstream joins and analysis.

SQL Queries

SQL queries were used to explore the dataset and generate analytical insights prior to building the dashboard.

Key analytical queries included:

- Category sales analysis
- Regional sales analysis
- Customer revenue contribution
- Product profitability analysis
- Discount impact analysis
- Monthly sales trend analysis

Query 1

```
SELECT
o.Order_ID,
o.Order_Date,
c.Customer_Name
FROM dbo.Orders o
JOIN dbo.Customers c
ON o.Customer_ID = c.Customer_ID;
```

This query returns a dataset showing which customer placed each order by joining the Customers and Orders tables.

	Order_ID	Order_Date	Customer_Name
1	CA-2014-128055	2014-03-31	Alex Avila
2	CA-2014-138100	2014-09-15	Alex Avila
3	CA-2015-121391	2015-10-04	Alex Avila
4	CA-2016-103982	2016-03-03	Alex Avila
5	CA-2017-147039	2017-06-29	Alex Avila
6	CA-2017-100230	2017-12-11	Allen Armold
7	CA-2016-126613	2016-07-10	Allen Armold
8	CA-2016-131065	2016-11-14	Allen Armold
9	CA-2015-140921	2015-02-03	Allen Armold
10	CA-2015-114503	2015-11-13	Allen Armold

Query 2

```
SELECT
o.Order_ID,
c.Customer_Name,
oi.Product_ID,
oi.Quantity,
oi.Sales
FROM dbo.Orders o
JOIN dbo.Customers c
ON o.Customer_ID = c.Customer_ID
```

```
JOIN dbo.Order_Items oi
ON o.Order_ID = oi.Order_ID;
```

Produces a dataset showing:

- which **customer placed an order**
- which **products were purchased**
- how many were purchased
- the **sales value of each item**

Each row represents **one product within an order**.

	Order_ID	Customer_Name	Product_ID	Quantity	Sales
1	CA-2016-152156	Claire Gute	FUR-BO-10001798	2	261.96
2	CA-2016-152156	Claire Gute	FUR-CH-10000454	3	731.94
3	CA-2016-138688	Darrin Van Huff	OFF-LA-10000240	2	14.62
4	US-2015-108966	Sean O'Donnell	FUR-TA-10000577	5	957.58
5	US-2015-108966	Sean O'Donnell	OFF-ST-10000760	2	22.37
6	CA-2014-115812	Brosina Hoffman	FUR-FU-10001487	7	48.86
7	CA-2014-115812	Brosina Hoffman	OFF-AR-10002833	4	7.28
8	CA-2014-115812	Brosina Hoffman	TEC-PH-10002275	6	907.15
9	CA-2014-115812	Brosina Hoffman	OFF-BI-10003910	3	18.50
10	CA-2014-115812	Brosina Hoffman	OFF-AP-10002892	5	114.90

Query 3

```
SELECT
    o.Order_ID,
    o.Order_Date,
    c.Customer_Name,
    p.Product_Name,
    p.Category,
    oi.Quantity,
    oi.Sales,
    oi.Profit
FROM dbo.Order_Items oi
JOIN dbo.Orders o
    ON oi.Order_ID = o.Order_ID
JOIN dbo.Customers c
    ON o.Customer_ID = c.Customer_ID
JOIN dbo.Products p
    ON oi.Product_ID = p.Product_ID;
```

Creates a combined dataset showing:

- who bought the product
- what product was purchased
- when the order occurred
- how many units were sold

- the sales and profit generated.

Each row represents **one product within an order**.

	Order_ID	Order_Date	Customer_Name	Product_Name	Category	Quantity	Sales	Profit
1	CA-2016-152156	2016-11-08	Claire Gute	Bush Somerset Collection Bookcase	Furniture	2	261.96	41.91
2	CA-2016-152156	2016-11-08	Claire Gute	Hon Deluxe Fabric Upholstered Stacking Chairs, Ro...	Furniture	3	731.94	219.58
3	CA-2016-138688	2016-06-12	Darrin Van Huff	Self-Adhesive Address Labels for Typewriters by Univ...	Office Supplies	2	14.62	6.87
4	US-2015-108966	2015-10-11	Sean O'Donnell	Bretford CR4500 Series Slim Rectangular Table	Furniture	5	957.58	-383.03
5	US-2015-108966	2015-10-11	Sean O'Donnell	Eldon Fold 'N Roll Cart System	Office Supplies	2	22.37	2.52
6	CA-2014-115812	2014-06-09	Brosina Hoffman	Eldon Expressions Wood and Plastic Desk Accessor...	Furniture	7	48.86	14.17
7	CA-2014-115812	2014-06-09	Brosina Hoffman	Newell 322	Office Supplies	4	7.28	1.97
8	CA-2014-115812	2014-06-09	Brosina Hoffman	Mitel 5320 IP Phone VoIP phone	Technology	6	907.15	90.72
9	CA-2014-115812	2014-06-09	Brosina Hoffman	DXL Angle-View Binders with Locking Rings by Sam...	Office Supplies	3	18.50	5.78
10	CA-2014-115812	2014-06-09	Brosina Hoffman	Belkin F5C206VTEL 6 Outlet Surge	Office Supplies	5	114.90	34.47

Query 4

```sql
SELECT
    p.Category,
    SUM(oi.Sales) AS Total_Sales,
    SUM(oi.Profit) AS Total_Profit
FROM dbo.Order_Items oi
JOIN dbo.Products p
    ON oi.Product_ID = p.Product_ID
GROUP BY p.Category
ORDER BY Total_Profit DESC;
```

This query groups the total sales and profit by product category. As the results show, technology products made the most sales and profit whereas office supplies had the least sales, but furniture had the least profit

	Category	Total_Sales	Total_Profit
1	Technology	836154.04	145455.35
2	Office Supplies	719046.93	122490.08
3	Furniture	741999.79	18451.13

Query 5

```sql
SELECT
    o.Region,
    SUM(oi.Sales) AS Total_Sales,
    SUM(oi.Profit) AS Total_Profit
FROM dbo.Order_Items oi
JOIN dbo.Orders o
    ON oi.Order_ID = o.Order_ID
GROUP BY o.Region
ORDER BY Total_Sales DESC;
```

This query groups the total sales and profit by region. As the results show, the West region had the most sales and maximum profit while South region had the least sales while Central had the least profit

Results | Messages

	Region	Total_Sales	Total_Profit
1	West	725457.84	108418.32
2	East	678781.28	91522.50
3	Central	501239.78	39706.26
4	South	391721.86	46749.48

Query 6

```
SELECT TOP 10
    c.Customer_Name,
    SUM(oi.Sales) AS Total_Sales
FROM dbo.Order_Items oi
JOIN dbo.Orders o
    ON oi.Order_ID = o.Order_ID
JOIN dbo.Customers c
    ON o.Customer_ID = c.Customer_ID
GROUP BY c.Customer_Name
ORDER BY Total_Sales DESC;
```

This query shows the top 10 customers by sales

	Customer_Name	Total_Sales
1	Sean Miller	25043.07
2	Tamara Chand	19052.22
3	Raymond Buch	15117.35
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.57
6	Ken Lonsdale	14175.23
7	Sanjit Chand	14142.34
8	Hunter Lopez	12873.30
9	Sanjit Engle	12209.44
10	Christopher Conant	12129.08

Query 7

```
SELECT
    p.Sub_Category,
    SUM(oi.Sales) AS Total_Sales,
    SUM(oi.Profit) AS Total_Profit
FROM dbo.Order_Items oi
JOIN dbo.Products p
    ON oi.Product_ID = p.Product_ID
GROUP BY p.Sub_Category
ORDER BY Total_Profit DESC;
```

This query shows which product sub-categories generate the most profit and which ones generate low or negative profit. This can reveal unprofitable product lines.

Results | Messages

	Sub_Category	Total_Sales	Total_Profit
1	Copiers	149528.01	55617.85
2	Phones	330007.10	44516.04
3	Accessories	167380.31	41936.73
4	Paper	78479.24	34053.16
5	Binders	203412.71	30221.42
6	Chairs	328449.09	26590.12
7	Storage	223843.59	21278.96
8	Appliances	107532.14	18138.00
9	Furnishings	91705.12	13059.18
10	Envelopes	16476.38	6964.03
11	Art	27118.80	6527.83
12	Labels	12486.30	5546.18
13	Machines	189238.62	3384.73
14	Fasteners	3024.25	949.52
15	Supplies	46673.52	-1189.02
16	Bookcases	114880.00	-3472.59
17	Tables	206965.58	-17725.58

Query 8

```sql
SELECT
    oi.Discount,
    SUM(oi.Sales) AS Total_Sales,
    SUM(oi.Profit) AS Total_Profit
FROM dbo.Order_Items oi
GROUP BY oi.Discount
ORDER BY oi.Discount;
```

This query shows how different discount levels affect total sales and profit, helping identify whether higher discounts reduce profitability.

	Discount	Total_Sales	Total_Profit
1	0.0000	1087908.47	320987.12
2	0.1000	54369.30	9029.20
3	0.1500	27558.55	1418.97
4	0.2000	764594.28	90337.58
5	0.3000	103226.68	-10369.34
6	0.3200	14493.45	-2391.16
7	0.4000	116417.83	-23057.07
8	0.4500	5484.98	-2493.12
9	0.5000	58918.57	-20506.48
10	0.6000	6644.68	-5944.63
11	0.7000	40620.29	-40075.40
12	0.8000	16963.68	-30539.11

Query 9

```sql
SELECT
    YEAR(o.Order_Date) AS Order_Year,
    MONTH(o.Order_Date) AS Order_Month,
    SUM(oi.Sales) AS Total_Sales
FROM dbo.Order_Items oi
JOIN dbo.Orders o
    ON oi.Order_ID = o.Order_ID
GROUP BY
    YEAR(o.Order_Date),
    MONTH(o.Order_Date)
ORDER BY
    Order_Year,
    Order_Month;
```

This query shows total sales by month

	Order_Year	Order_Month	Total_Sales
1	2014	1	14236.90
2	2014	2	4519.92
3	2014	3	55691.04
4	2014	4	28295.35
5	2014	5	23648.26
6	2014	6	34595.12
7	2014	7	33946.37
8	2014	8	27909.47
9	2014	9	81777.32
10	2014	10	31453.37
11	2014	11	78628.73
12	2014	12	69545.63
13	2015	1	18174.08
14	2015	2	11951.40
15	2015	3	38726.26
16	2015	4	34195.25

Query 10

```sql
SELECT TOP 10
    p.Product_Name,
    SUM(oi.Sales) AS Total_Sales
FROM dbo.Order_Items oi
JOIN dbo.Products p
    ON oi.Product_ID = p.Product_ID
GROUP BY p.Product_Name
ORDER BY Total_Sales DESC;
```

This query shows the top 10 products by sales

	Product_Name	Total_Sales
1	Canon imageCLASS 2200 Advanced Copier	61599.83
2	Fellowes PB500 Electric Punch Plastic Comb Binding ...	27453.38
3	Cisco TelePresence System EX90 Videoconferencing ...	22638.48
4	HON 5400 Series Task Chairs for Big and Tall	21870.57
5	GBC DocuBind TL300 Electric Binding System	19823.47
6	GBC Ibimaster 500 Manual ProClick Binding System	19024.50
7	Hewlett Packard LaserJet 3310 Copier	18839.68
8	HP Designjet T520 Inkjet Large Format Printer - 24" C...	18374.90
9	GBC DocuBind P400 Electric Binding System	17965.07
10	High Speed Automatic Electric Letter Opener	17030.31

Key Performance Metrics

The following key performance indicators were used to evaluate business performance.

Total Sales
Total revenue generated from all product transactions.

Total Profit
Net profit generated after accounting for discounts and costs.

Total Quantity
Total number of product units sold across all orders.

Profitability by Category
Measures which product categories contribute the most to overall profit.

Regional Sales Performance
Evaluates how revenue and profitability vary across geographic regions.

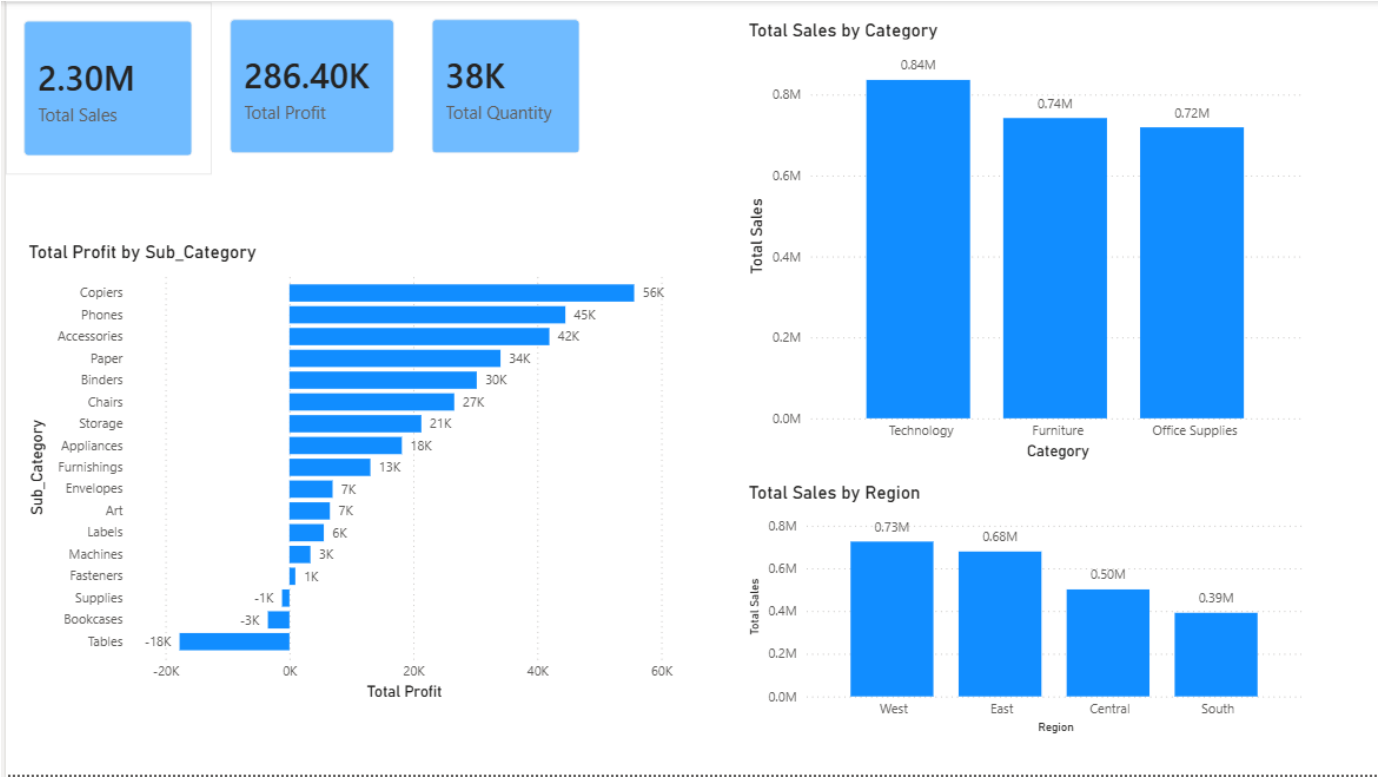
Discount Impact
Examines how different discount levels influence profitability.

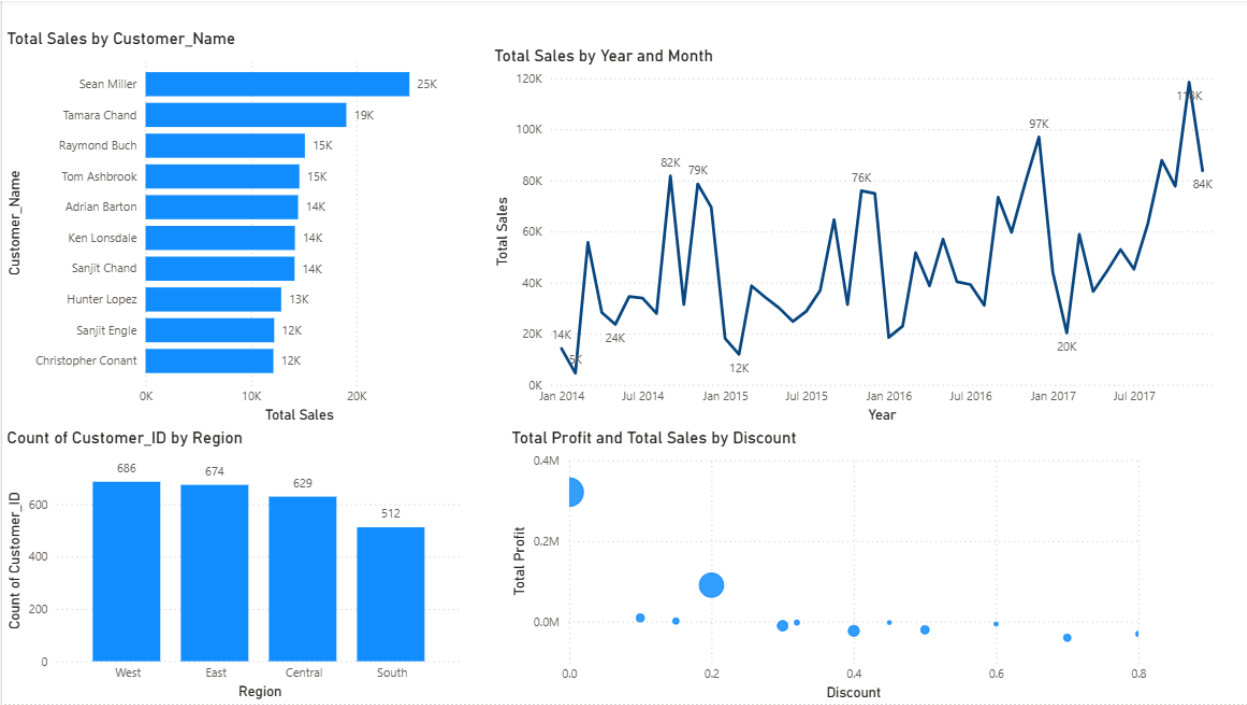
Power BI Dashboard

Power BI was used to build an interactive dashboard that visualises key performance indicators and analytical insights derived from the SQL analysis.

The dashboard includes:

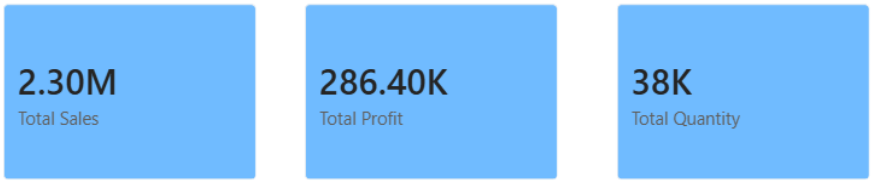
- KPI cards summarising total sales, profit, and quantity sold
- Product category performance
- Regional revenue comparisons
- Product profitability analysis
- Customer sales contribution
- Monthly sales trends
- Discount impact on profitability





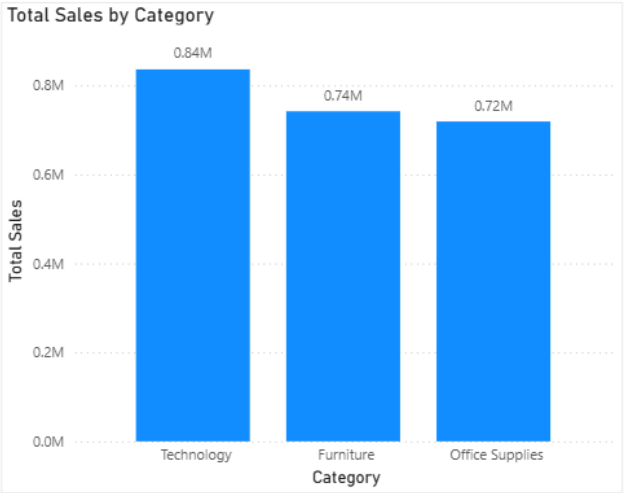
Dashboard Visualisations

Total KPIs



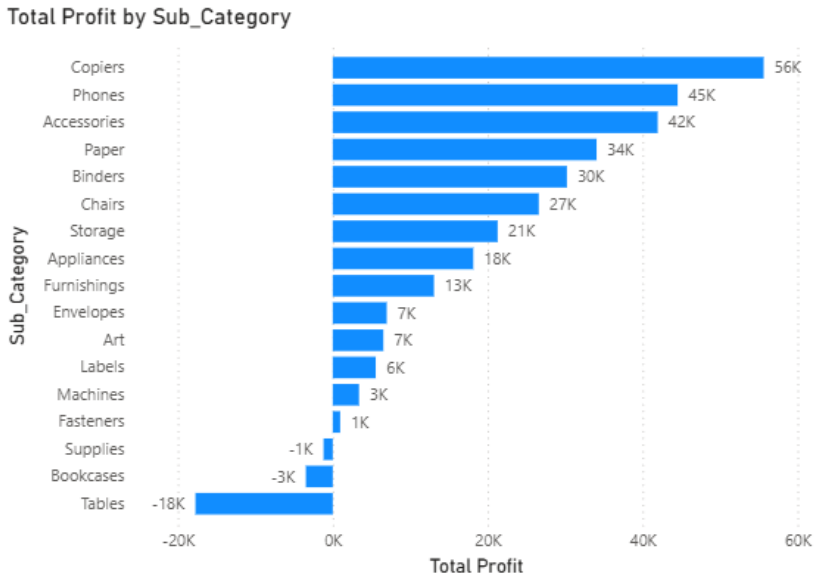
These are the figures generated for the KPIs

Sales by Category



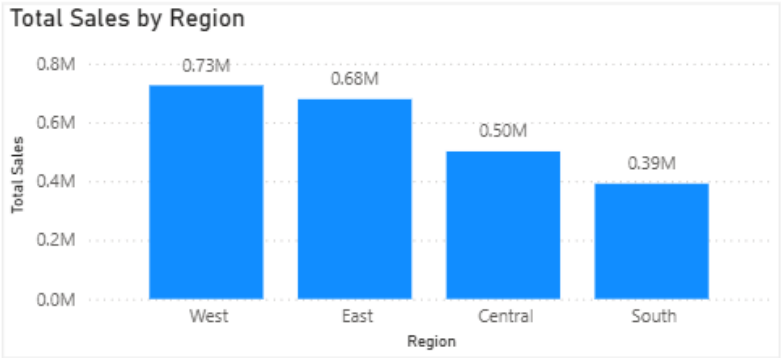
This chart shows total revenue generated by each product category.
Technology generates the highest sales, followed by Furniture and Office Supplies.
This suggests technology products are the primary revenue driver for the business.

Profit by Sub-Category



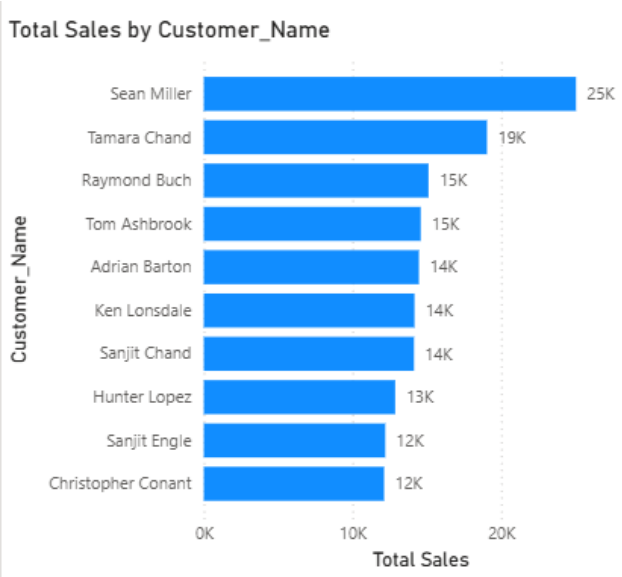
This visual analyses profitability at a more detailed product level.
While several sub-categories generate strong profits, Tables and Bookcases show negative profitability.
This may indicate high discounting or cost issues within these product groups.

Sales by Region



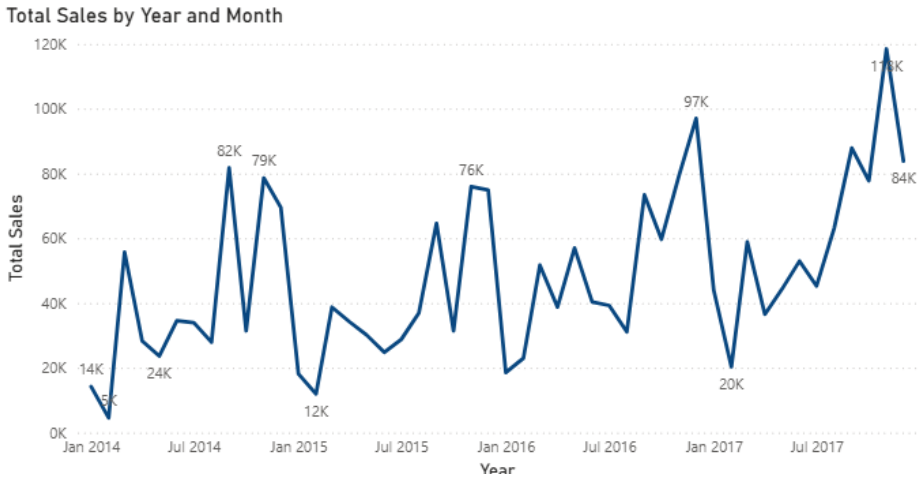
This chart compares revenue performance across regions.
The West region generates the highest total sales, followed by East, Central, and South.
This suggests stronger market performance in western regions.

Top Customers by Sales



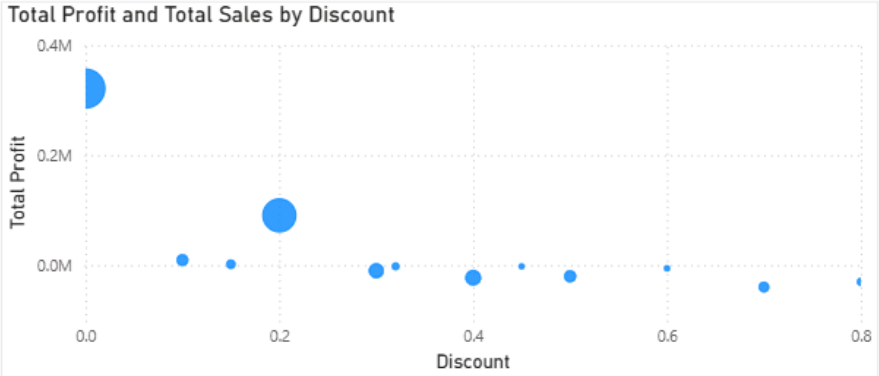
This visual identifies the top customers contributing the most revenue.
These customers represent key accounts and may be important for retention strategies.

Monthly Sales Trend



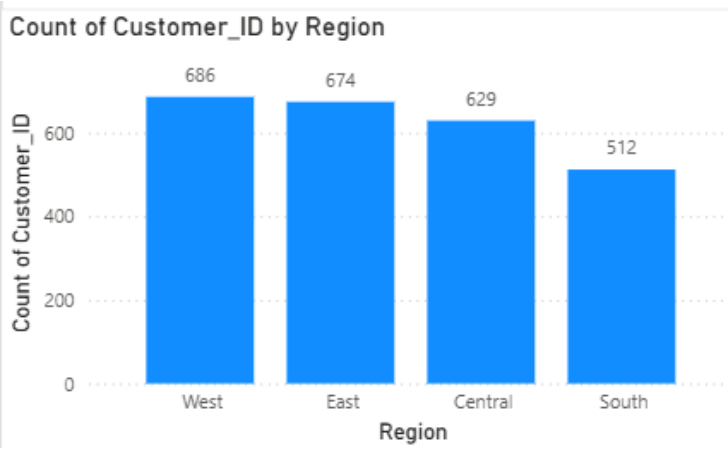
This line chart shows sales performance over time.
The trend indicates overall growth with several seasonal spikes in demand.

Discount vs Profit



This scatter plot explores the relationship between discount levels and profitability.
Higher discounts are generally associated with lower or negative profit values.

Count of Unique Customers by Region



This graph shows the number of different customers divided by region.

Key Business Insights

1. Technology products generate the highest revenue and profit, making them the most important category for the business.
2. Certain product sub-categories such as Tables and Bookcases produce negative profitability, indicating potential pricing or cost issues.
3. The West region contributes the highest overall sales, suggesting stronger demand or market penetration in this region.
4. Sales show a gradual upward trend over time, indicating business growth with periodic seasonal fluctuations.
5. Higher discount levels are associated with declining profitability, suggesting aggressive discounting strategies may be eroding margins.

Business Recommendations

Based on the analysis, the following actions are recommended:

1. **Review pricing strategies for low-profit sub-categories**
   Product groups such as Tables and Bookcases should be evaluated for pricing, cost structure, or discount policies to improve profitability.
2. **Optimise discount strategies**
   High discount levels appear to significantly reduce profit margins. Discount policies should be reviewed to balance revenue growth with profitability.

3. **Focus marketing efforts on high-performing regions**
   The West region demonstrates strong revenue performance and may present opportunities for further expansion.

4. **Strengthen relationships with top customers**
   High-value customers identified in the analysis should be prioritised through loyalty programs and targeted marketing initiatives.

5. **Expand high-performing product categories**
   Technology products generate the strongest revenue performance and could be further leveraged through expanded offerings or marketing campaigns.

Limitations of the Analysis

This analysis is based on a historical dataset and assumes the underlying data accurately reflects operational performance. The dataset does not include operational cost breakdowns or supply chain information, which may further influence profitability outcomes.

Future analysis could incorporate additional datasets such as inventory costs, supplier pricing, or marketing expenditure to provide a more comprehensive profitability model.

Project Outcomes/ Skills Demonstrated

This project demonstrates the application of data analytics techniques to transform raw transactional data into actionable business insights.

Through this analysis, the following skills were applied:

Data Cleaning and Validation

- Identification and correction of missing date values
- Resolution of datatype inconsistencies affecting financial data

Relational Data Modelling

- Normalisation of the dataset into four structured tables
- Implementation of primary and foreign key relationships to maintain data integrity

SQL Data Analysis

- Use of joins to integrate multiple datasets
- Aggregation queries to analyse sales, profit, and customer performance

- Identification of profitability trends across products, regions, and discount levels

Data Visualisation

- Development of an interactive Power BI dashboard to communicate key insights
- Creation of KPI indicators and analytical visuals to highlight revenue drivers and profitability risks

Business Insight Generation

- Identification of high-performing product categories and regions
- Detection of unprofitable product sub-categories
- Evaluation of the impact of discount strategies on profitability

Conclusion

This project demonstrates how SQL and Power BI can be combined to transform raw transactional data into actionable business insights. Through structured data cleaning, relational modelling, and analytical querying, key profitability drivers were identified across product categories, regions, and customer segments.

The resulting Power BI dashboard provides an interactive tool for monitoring sales performance and identifying areas where profitability can be improved.